

WIZ: DreamWorks GPU-Accelerated Interactive Hair/Fur Deformation and Visualization Toolset

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Figure 1: Hair and fur of characters from the DreamWorks film *The Bad Guys* visualized using WIZ in the 3-D viewport.

ABSTRACT

WIZ is a hair/fur visualization toolset at DreamWorks that uses the power of the GPU for both interactive deformation and display in the 3-D viewport. WIZ closely approximates the final render hair motion on the GPU based on skin and guide curve motion, and provides the artist with various options for render hair deformation. It can use the character's hair textures and closely approximate the final look of the hair. Hundreds of thousands or millions of curves in motion can be visualized close to real-time which can help the artist evaluate hair/fur motion iteratively in a shot without having to do a render or having to use much slower existing viewport displays, thus providing significant time and resource savings.

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1 INTRODUCTION

Hair/fur can consist of hundreds of thousands or millions of render curves that need to be deformed and visualized for iterative shot performance evaluation. Using just the CPU for this process is slow even with multi-threading. Waiting for full renders is even slower and can hold up renderfarm resources. Therefore, artists typically evaluate hair/fur motion in the 3-D viewport by visualizing just guide curves as curves or tubes. WIZ tackles this issue of dealing with huge amounts of data by using the power of the GPU (thousands of cores) to both deform the render curves and display them with shading and lighting interactively in the 3-D viewport.

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2 LIVE STYLERS VS POST-STYLER MOTION

One option in using the GPU for generating render curves is to implement the interpolation between the guide curves and all the render curve stylers (such as kink, curl, clump, etc.) in the GPU (in addition to the CPU). This would keep the stylers live during a shot's hair motion evaluation and accurate to the final CPU render. However, there could be a lot of stylers, some involving complex algorithms which could be difficult to implement on the GPU. Newer stylers would need a GPU and a CPU version and be kept in sync. Moreover, in a groom setup, several stylers could be stacked one atop another (including dependencies), slowing down their overall GPU evaluation making it non-interactive. Also, grooms could come in baked with final render curves from third-party packages, where stylers used to create that groom are not available.

Taking these factors into consideration, WIZ uses the approach of deforming the final baked render curves, created at reference pose, using skin and guide curves' motion at any shot frame. While this post-styler deformation approach is an approximation, it avoids slower and more complex styler evaluation, and is visually close enough for the artist to evaluate full render hair motion iteratively and interactively in the 3-D viewport during shot work.

3 WIZ CACHE

In order to deform and display the render hairs using the GPU, WIZ first creates an Alembic cache file per hairstyle one time on disk using the CPU. This WIZ Cache stores all render curves' reference pose data such as point position, point width, point color, skin bind data, and guide curves' bind data per render curve or render curve point. The skin bind data consists of the polygon of the skin mesh and the corresponding parametric UV of the point on that polygon that the curve's root point is closest to, and the local offset of the render curve points about that closest point on the skin. The guide curves' bind data consists of the guide curves, the interpolation weights of those guide curves, the parametric UV of the closest point on the purely guide-interpolated curve from a render curve point, and the corresponding closest distance, D_{cl} . The size of a WIZ Cache for about 5 million curve points is ≈ 220 MB on disk.

4 WIZ GPU DEFORMATION

WIZ provides several modes of deformation of the render curves based on skin motion and guide curves' motion. In addition to the guide curves' animated positions, the reference guide curves are root attached to the animated skin and their positions are passed to WIZ as point attributes, namely P_{guide_orig} . WIZ uses the bind data information stored in the WIZ Cache based on artist-selected deformation modes. The deformation operation of WIZ is on the GPU using a GLSL Vertex Shader. The deformation modes are:

- **Only Skin Attachment:** Using the skin bind data stored in the WIZ Cache at reference pose, the render curves at reference pose are deformed to the animated skin pose using a rigid-root attachment. This mode could be used in cases where there is no guide curve motion but just skin motion.
- **Use Guide Motion - Interpolation:** Using the guide curves' bind data in the WIZ Cache, the render curves are created by purely interpolating between the corresponding guide curves' at the animated pose. In this mode, however all styling is lost. This mode could be used for smoother hairstyles or for hairstyles with less styling. The interpolation technique employed here is not a simple corresponding guide point (control vertex) interpolation. During the WIZ Cache creation stage, at the reference pose, the parametric UV of the closest point from each render curve point to each corresponding guide curve is stored. The interpolation of the render curve happens between those points (calculated using that parametric UV at the animated guide pose), and not just the guide control vertex positions. This helps to closely approximate the length of the render curves, which could have been curtailed by length-based stylers that could have taken effect after the interpolation step in the original groom.
- **Use Guide Motion - Position Wrap:** In this mode, for every render curve, first an interpolated curve's positions, P_{interp_orig} , are obtained using the P_{guide_orig} attribute of the corresponding guide curves. Next, an interpolated curve's positions, P_{interp_anim} , are obtained using the animated positions of the corresponding guide curves. Then, using skin bind attachment the reference render curve from the WIZ Cache is rigid-root deformed to the animated skin pose to obtain a pose with positions $P_{r_skin_bind}$. The final position of the render curve P_r is calculated by adding the difference between P_{interp_anim} and P_{interp_orig} to $P_{r_skin_bind}$, namely $P_r = P_{r_skin_bind} + (P_{interp_anim} - P_{interp_orig})$. The skin binding and interpolation steps execute similar to the prior two modes.
- **Use Guide Motion - Distance Wrap:** In this mode, first curve positions $P_{r_skin_bind}$ and P_{interp_anim} are calculated as in the prior mode. Then the distance between corresponding points of those two positions is kept to be the distance D_{cl} (stored in WIZ Cache), i.e., $P_r = P_{interp_anim} + D_{cl} * (V_n / ||V_n||)$, where $V_n = (P_{r_skin_bind} - P_{interp_anim})$.

Such high-fidelity approaches at render curve density increase the accuracy of the approximation, and are possible and interactive because of GPU usage. Currently, the above deformation modes have been producing good results and the need for a complete wrap has not presented itself. A complete wrap involves local axis calculation at render curve density which could slow down execution and also introduce twisting artifacts based on the groom's nature.



Figure 2: Visualizing the cheek fur puff effect using guide curves in the 3-D viewport [left], WIZ in the 3-D viewport [middle], and final lighting rendered image [right].

5 WIZ GPU DISPLAY

First, the render curves' point positions are calculated using a GLSL Vertex Shader. Then, a GLSL Geometry Shader converts the curves into polygonal strips facing the camera. The widths of the strips and the color are obtained from the data stored in the WIZ Cache. It should be noted here that since the point colors are stored in the WIZ Cache, the artist can also assign point colors other than the material colors such as clump colors, or guide curve association colors in the WIZ Cache for debugging or for a better technical read (including breakdown of regions and their motion) of the entire set of render curves interactively. Finally, the geometry is lit and shaded in a GLSL Fragment Shader and responds interactively to scene lighting. The artist can control the shading with hair material properties such as ambient, diffuse, specular, and shininess. Level of Detail controls such as *Density* and *Radius Scale* are also provided which can be controlled by the artist based on screen size and/or character count to speed up the display further in a shot.

6 IMPLEMENTATION

WIZ is implemented as custom nodes in a third-party package. The *WIZ Cache* node creates the cache one time per hair style. The *WIZ Bind* node calculates all the skin binding data and curve binding data on the CPU at reference pose needed to create the cache. The *WIZ Display* node performs both the deformation and display of animated render hair/fur on the GPU. The WIZ display mode is included in the hair shot pipeline along with lower-fidelity or slower display modes such as displaying guide curves [Figure 2], tubes, or a full CPU evaluation of the deformation. The CPU code is written in C++, and the GPU code is written using GLSL shaders.

7 RESULTS AND CONCLUSION

WIZ provides the solution for interactive full density render hair/fur motion display for iterative shot work. WIZ uses the power of GPU for both interactive hair deformation and display in the 3-D viewport. Character hair motion can be visualized at greater than 85% accuracy in a 1920x816 resolution 3-D viewport at rates ranging from 15 to 70 frames per second (15 fps [5 million points / 60% screen space occupied by hair] to 70 fps [2.5 million points / 10% screen space]). This is much faster than CPU evaluation which can take 2 to 50 seconds per frame to generate and display (which is prohibitive for iterative shot work). Also, to compare, full renders can take minutes or hours to complete holding up renderfarm resources. Using WIZ, the *Character Effects* artists save significant time per frame, motion iteration, asset, and shot. WIZ was deployed on the film *The Bad Guys* and is continuing to be used in current films.